

L3/Event Analyzer Methodology

How the current model reads L3 and Events files, fixes timestamps, classifies technology, detects call sessions, separates connected, not connected, and dropped calls, and calculates setup time and duration.

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1. Input Data

- The analyzer consumes two families of records: decoded L3 signaling rows and device or modem Event rows.
- L3 rows provide protocol messages such as RRC, NAS, IMS SIP, attach, registration, handover, bearer, and release messages.
- Event rows provide call-state callbacks such as CALL_DIAL_INITIATED, CALL_DIALING, CALL_ALERTING, CALL_ACTIVE, CALL_DISCONNECTED, CallState, mPreciseCallState, and disconnect cause rows.
- Each row keeps source metadata, raw message text, decoded details, source file, row index, severity, latitude, longitude, and any parser metadata available from the uploaded archive.

2. Timeline Normalization

- Every L3 and Event row is merged into one chronological timeline in timelineBuilder.js.
- Timestamps can be epoch milliseconds, epoch seconds, full date strings, or time-only values such as 17:34:50.306.
- Time-only values are anchored to 1970-01-01 UTC so ordering stays stable and the visible time matches the log text.
- If raw logcat text contains a fresher embedded timestamp, the analyzer uses that embedded timestamp as the true event time.
- If an embedded event is stale compared with the CSV capture time, it is marked stale and is not allowed to create false call evidence.

3. Technology Classification

- Technology is inferred from procedure analysis, item protocol, interface names, source category, and raw message text.
- The normalization rule maps gsm or 2g to 2G, umts/utran/3g to 3G, lte/4g/enodeb/eps to LTE, and nr/5g/gnb to 5G.
- Protocol analysis also tags procedures using 3GPP families: LTE RRC, NR RRC, NAS, IMS, GSM RR/MM/CC, UMTS RRC, EPS, and NR NAS.
- For each call, the UI records technology at the start and at the end. If technology changes during mobility, the home table shows it as Start -> End.
- Unknown is kept when no reliable radio/protocol hint exists. The model does not invent a technology from unrelated text.

4. Procedure and Signaling Model

- protocolAnalyzer.js matches rows against a procedure library of known 3GPP messages and flows.
- Examples include GSM MO Call, UMTS CS MO Call, Paging, RRC Connection Establishment, Security Mode Control, Authentication, Attach, Registration, IMS SIP setup, and handover procedures.
- The procedure model records protocol, source node, destination node, specification family, success markers, failure markers, duration, and related evidence.
- signalingModel.js converts the merged timeline into the Excel View rows: Timestamp,

UE, eNB/gNB, MME/AMF/Core, Interface, Message, and raw Detail.

5. Call Session Detection

- sessionBuilder.js creates one call session when it sees a real start event.
- Outgoing starts are detected from CALL_DIAL_INITIATED, CALL_DIALING, or dial/calling/trying state.
- Incoming starts are detected only from CallState or mPreciseCallState rows that contain ringing or alerting plus incoming/MT-call wording.
- Generic ringing by itself is intentionally not a start because it created false calls in real traces.
- Duplicate start rows inside a one second window are deduplicated into the same session.
- A session ends on CALL_DISCONNECTED or an idle/disconnected/ended call-state row. Both terminal callbacks are captured because ordering matters.

6. Connection Evidence

- A call is connected only when strong connection evidence is found before the normalized end marker.
- Strong evidence includes SIP INVITE followed by SIP 200 OK followed by SIP ACK, explicit connected call state, non-remote callType=3 after alerting, or media/bearer establishment.
- Remote capability callType=3 is not treated as connection by itself. Local callType=3 can confirm the call after alerting.
- Codec evidence such as CODEC_AMR_NB, CODEC_AMR_WB, CODEC_EVS, or updateMediaCapabilities codec fields is kept as supporting evidence.
- Evidence after the actual end is ignored so delayed callbacks do not turn a failed attempt into a connected call.

7. Fixed-Duration Drive-Test Inference

- Some drive-test calls complete locally with weak explicit connection callbacks. For those, the analyzer has a conservative estimator.
- The estimator requires local cause 3, an attempt between 85 and 130 seconds, alerting evidence, and either codec evidence or modem disconnect before Idle.
- When those conditions match, connected time is inferred as end time minus 90 seconds.
- The inferred setup must be between 1 and 30 seconds and after alerting. Inferred values are marked with connectionEstimated.
- This rule is narrow by design because it should repair known scripted-call traces without hiding real setup failures.

8. Connected, Not Connected, Dropped

- callClassifier.js classifies a session after milestones and evidence are finalized.
- Connected means the call has connection evidence and no unrecovered abnormal termination evidence before the end.
- Not Connected means the session ended before connection evidence. Details can be Busy, Rejected, User Cancelled, Call Setup Failure, IMS Failure, Radio Failure, or Handover Failure.
- Dropped means the call connected first and then ended with abnormal release evidence such as radio failure, handover failure, IMS failure, bearer/media loss, SCG failure, or unrecovered RRC failure.
- Recovered RF problems are tracked. If RRC re-establishment or a recovery signal appears before release, that issue is exposed as evidence but does not force a dropped final result.

9. RF and Protocol Failure Signals

- RF/radio failure hints include radio link failure, RLF, RRC re-establishment failure, unexpected RRC release, bearer loss, lost signal, out of service, SCG fail, and SCG failure.
- Handover failures include handover or HO text combined with fail, failure, reject, drop, or timeout.
- IMS/SIP failures include IMS registration lost/deregistered/unregistered and SIP failure codes such as 408, 480, 486, 500, and 503 in SIP context.
- Bearer failures include voice bearer, media, RTP, or QoS flow loss/failure/release unexpected wording.
- Disconnect cause codes are mapped through disconnectCauseMapper.js and combined with L3 evidence to choose the most specific reason.

10. Duration Calculation

- durationCalculator.js uses the normalized first terminal marker as the end time, not later cleanup rows.
- Call setup time is dial/start to connected/setup completion. It stays null when the call never connected.
- Connected duration is connected time to normalized end time. It stays null for not-connected attempts.
- Attempt duration is dial/start to end time and is tracked separately from connected talk time.
- The summary uses connected media time as call duration, while still keeping total attempt duration for diagnostics.

11. Summary Output

- summaryBuilder.js counts total calls, connected, dropped, not connected, busy, rejected, setup failures, ongoing, and unknown.
- Average setup time is calculated from connected calls with positive setup time.
- Average talk time is calculated from connected duration values greater than zero.
- Success rate is connected calls divided by total calls.
- The home screen now shows the main cards plus a per-call table with call id, start, end, technology, result, setup time, duration, and reason.

12. Excel View and PDF Export

- Excel View is a signaling sheet, not a spreadsheet clone. It shows all filtered data in one scrollable table.
- The visible columns are Timestamp, UE, eNB/gNB, MME/AMF/Core, Interface, and Message.
- Interface and Message filters live in their column headers. UE, radio, and core columns support uplink/downlink filtering.
- The selected row detail panel shows the raw message first, then supporting decoded fields.
- Download Summary creates a PDF with Summary, Call Summary, Technology Summary, and Sheet Messages. Sheet Messages contain Timestamp, Interface, Message, and Detail/raw message.

13. Main Source Files

File Responsibility

```
timestampUtils.js embedded timestamp extraction and stale-event checks
timelineBuilder.js merge L3/Event rows into chronological timeline
protocolAnalyzer.js 3GPP procedure detection and technology/protocol context
sessionBuilder.js call session creation, milestones, SIP/media/RF evidence
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callClassifier.js connected/not connected/dropped decision and reason
durationCalculator.js setup, connected duration, and attempt duration
summaryBuilder.js aggregate call counters and averages
signalingModel.js Excel View signaling rows and lane/message model
pdfReport.js in-app PDF reports and sheet-message summary export
L3EventsTab.jsx home cards, per-call table, tabs, and export actions

14. Validation Dataset Results

- The corrected analyzer was checked against the uploaded archives used during development.

Archive Total Connected Not Connected Dropped

Log_20260807_094502.zip 4 2 1 1
BestL3.zip 5 5 0 0
CSTEST.zip 6 2 2 2

15. Practical Reading Guide

- When a count looks wrong, first inspect stale timestamp flags and whether an Event row was delayed by log buffering.
 - If a not-connected call appears connected, check whether the evidence came from remote capability callType=3 or from a row after end time.
 - If a connected call appears dropped, check whether RF failure evidence was recovered by RRC re-establishment or later normal SIP/BYE release.
 - If setup time is zero or missing, confirm that dial/start and connected evidence both have usable timestamps.
 - If technology is Unknown, inspect protocol, interface, source category, and raw text.
- The model only labels a technology when one of those inputs has a clear hint.